Faith Frings

PhD Candidate

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Education

Doctor of Philosophy, Integrative Biology *08/2022-present*

University of New Hampshire

- Advisor: Dr. Melissa Aikens | Dissertation: Building data literate biologists: Investigating student motivational dynamics in statistical programming across diverse biology courses

Master of Science, Biology with a concentration in ecology and evolution *08/2020-08/2022*

University of South Florida

- Advisor: Dr. Luanna Prevost | Thesis: Mixed-method analysis of undergraduate student interpretation of different phylogenetic tree styles

Bachelor of Science, Marine Biology *08/2017-05/2020*

University of South Florida

Research Experience

Dissertation (2022 - present)

Advised by Dr. Melissa Aikens

Characterizing student R programming motivation and engagement

- Conducted multilevel modeling to examine changes in students' motivation to learn R programming
- Conducted path analysis to investigate relationships among students' R programming mindsets, task values, and effort on R programming tasks

Examining student R programming motivation across course types

- Collected and analyzed open-response survey data to identify factors shaping students' motivation across R-integrated biology courses

Exploring student motivation to learn data science in the field

- Collected and analyzed semi-structured interviews to identify factors shaping students' motivation to learn R programming in a field-based data science course

Factors explaining student R programming resource use in biological research

- Collected and analyzed semi-structured interviews, participant observations, and surveys to examine how and why students select resources when programming in a biological research course

Thesis (2020 - 2022)

Advised by Dr. Luanna Prevost

Examining student phylogenetic tree interpretations across tree styles

- Conducted thematic analysis, phi correlations, and Mann-Whitney U tests of open-response data to characterize undergraduate students' reasoning across different phylogenetic tree styles

Undergraduate Research (2019 - 2020)

Advised by Dr. Luanna Prevost

Student reasoning about food webs

- Conducted thematic analysis and phi correlations of open-response data examining students' interpretations of food webs

Graduate Research Assistant

Biology Education Research Group (BERG), UNH

Community college transfer student persistence *2022-2026*

- Conducted semi-structured interviews and thematic analysis to investigate supports and barriers experienced by community college transfer students at a four-year institution

Biology Education Research Group (BERG), UNH

STUDENT GRAPHICAL REASONING *2022-2026*

- Conducted thematic analysis of open-response data to characterize students' graphical reasoning

Prevost Lab, USF

AMGEN BIOTECHNOLOGY EDUCATION PROJECT *2022*

- Developed online instructional resources for biology educators through an Amgen Foundation-funded biotechnology education project

Teaching Experience

PRIMARY INSTRUCTOR

BIOSM 2350/MEFB 633: R by the Sea: Data Analysis in Marine Sciences *06/2026*

UNDERGRADUATE, 1 SUMMER TERM

GRADUATE TEACHING ASSISTANT

BIOL 706/806: Data Science for Life Sciences *2023 - 2026*

UNDERGRADUATE/GRADUATE, 4 SEMESTERS

BIOL 528: Applied Biostatistics *2023 - 2026*

UNDERGRADUATE, 4 SEMESTERS

BIOSM 3290/MEFB 714: Field Animal Behavior *2025*

UNDERGRADUATE, 1 SUMMER TERM

BIOSM 4990/MEFB 751: Research in Biology *2024*

UNDERGRADUATE, 1 SUMMER TERM

BIOSM 1610/MEFB 674: Ecology & the Marine Environment *2024*

UNDERGRADUATE, 1 SUMMER TERM

BIOL 412: Introductory Biology Laboratory *2022*

UNDERGRADUATE, 1 SEMESTER

BSC 2011: Biology II – Biological Diversity *2022*

UNDERGRADUATE, 1 SEMESTER

BSC 2011L: Biology II – Biological Diversity Laboratory *2020 - 2021*

UNDERGRADUATE, 2 SEMESTERS

BSC 1005: Biological Principles for Non-Majors *2021*

UNDERGRADUATE, 1 SEMESTER

WORKSHOPS

R by the Sea - Japan. Advanced Institute for Marine Ecosystem Change (WPI-AIMEC), Tohoku University. *01/2026*

CO-TAUGHT A FIVE-DAY INTRODUCTORY R AND STATISTICS WORKSHOP

INVITED LECTURES

1. **Frings, F.**, & Villeneuve, D. (2025, September). *Graduate school advice & individual development plans*. Introductory Graduate Seminar (BIOL 901). University of New Hampshire.

INVITED FACILITATOR

Professional Development in Evidence-Based Teaching Series (PDET) *2026*

Teaching Distinctions & Professional Development

UNH Graduate Student Excellence in Teaching Award *2025*

RECOGNITION BY THE DEPARTMENT OF BIOLOGICAL SCIENCES FOR TEACHING ACHIEVEMENT

Human Subjects Research Certificate *2019,22,25*

WEB-BASED COURSE MODULES ON RESPONSIBLE CONDUCT OF HUMAN SUBJECTS RESEARCH

Journal Articles

2. **Frings, F.**, Heim, A., & Prevost, L. (in preparation). *Mixed-method analysis of undergraduate student interpretation of different phylogenetic tree styles*.

1. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (in review). *Preparing data-literate biologists: Examining motivational factors that drive student effort in R programming*.

Book Chapters

1. Esparza, D., Amin, N. E., **Frings, F.**, Halmo, S. M., Hazlett, Z. S., Heim, A. B., Uminski, C., Von der Mehden, B., Wright, A. M., & Yang, M. (accepted). *Guiding discipline-based education research trainees through career transitions with inclusive and accessible organizational support* (C. Etson, T. Lightner, & R. Greenler, Eds.; 1st ed.). Lexington Books.

Media

1. **Frings, F.**, Levine, A., Madrigal, L., Plank, L., & Richards, C. (2021). *Celebrating the importance of evolution*. Frontiers for Young Minds.

Presentations

Underlined names indicate undergraduate research mentees.

15. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (2026, July). *Preparing data literate biologists: What drives student effort in R programming?* [Talk]. The Society for the Advancement of Biology Education Research 16th Annual Meeting. East Lansing, MI, USA.

14. Koopman, C., Greene, A., **Frings, F.**, & Aikens, M. L. (2026, April). *Motivating factors for students learning R programming in STEM courses*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.

13. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (2026, February). *Preparing data literate biologists: What drives student effort in R programming?* [Talk]. University of New Hampshire, Department of Biological Sciences Spring Seminar Series.

12. **Frings, F.**, & Aikens, M. L. (2025, July). *Exploring student motivation in programming: Insights from a hybrid field-based data science course*. [Poster]. The Society for the Advancement of Biology Education Research 15th Annual Meeting. Minneapolis, MN, USA.

11. Dearth, A., **Frings, F.**, & Aikens, M. L. (2025, April). *Investigating student engagement in statistical programming in biology*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.

10. **Frings, F.**, & Aikens, M. L. (2025, April). *Characterizing the biology students' mindsets and task values towards programming*. [Talk]. The University of New Hampshire Annual Graduate Research Conference.

9. Maddox, M., Prieto, A., **Frings, F.**, & Aikens, M. L. (2025, April). *Exploring students' programming ability beliefs in a data science course*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.

8. **Frings, F.**, & Aikens, M. L. (2025, February). *Characterizing the biology students' mindsets and task values towards programming*. [Talk]. University of New Hampshire, Department of Biological Sciences' Spring Seminar Series.

7. **Frings, F.**, & Aikens, M. L. (2024, July). *Community college transfer students' perceptions of supports and barriers to persistence in their major at a four-year institution*. [Poster]. The Society for the Advancement of Biology Education Research 14th Annual Meeting. Minneapolis, MN, USA.

6. Amovic, S., Maddox, M., **Frings, F.**, Schuchardt, A., & Aikens, M. L. (2024, April). *Analysis of biology students' cognitive challenges when learning statistics in a course-based undergraduate research experience*. [Poster]. The

University of New Hampshire Annual Undergraduate Research Conference.

5. Weiss, V., **Frings, F.**, & Aikens, M. L. (2023, April). *Supports and barriers affecting community college transfer student degree persistence at a four-year institution*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.

4. **Frings, F.**, & Prevost, L. P. (2022, October). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Talk]. University of New Hampshire, Department of Biological Sciences Fall Seminar Series.

3. **Frings, F.**, & Prevost, L. P. (2022, March). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Poster]. The 4th Biennial CIRCLE Research Conference. St. Louis, MO, USA.

2. **Frings, F.**, & Prevost, L. P. (2021, March). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Poster]. The 113th Sigma Xi Annual Meeting & Student Research Conference. Virtual Conference.

1. **Frings, F.**, Grissett, C., & Prevost, L. P. (2020, April). *Correlation between competition and arrow reversals in students' responses to conceptual food web questions*. [Poster]. The University of South Florida 18th Annual Undergraduate Research Conference. Virtual Conference.

Research Mentoring

Undergraduate Mentees

Ashley Greene — *2026*

Carleigh Koopman — *2025-2026*

Abigail Dearth — *2025*

Ava Prieto — *2025*

Sara Amovic — *2023-2024*

MeiLing Maddox — *2023-2025*

Virginia Weiss — *2022 - 2023*

Awards & Fellowships

Student Teaching Award Fellowship — *2025*
UNH Graduate School, University of New Hampshire — *$2000*

Bill Woods Graduate Student Poster Award — *2024*
Society for the Advancment of Biology Education Research annual meeting — *$500*

Travel Award — *2024, 2025*
Department of Biological Sciences, University of New Hampshire — *$800*

Travel Award — *2024, 2026*
UNH Graduate School, University of New Hampshire — *$800*

S-STEM Scholarship — *2020-2021*
Integrative Biology Department, University of South Florida — *$5,600*

Grants in Support of Research

Graduate Student Research Fund — *2025*
School of Marine Sciences and Ocean Enginnering, University of New Hampshire — *$998*

Professional Service

2023-present

UNH R Users Group Organizer

- Organize and lead weekly R Users Group meetings

2025 - 2026

SABER Virtual Day Conference Committee

- Co-developed and co-hosted the inaugural Virtual Day of the 16th SABER annual meeting

2025-2026

SABER Discipline-Based Education Research-Scholars in Training (DBER-SiT) Chair

- Organized and delegated DBER-SiT committee projects

2024-2025

SABER DBER-SiT Chair-elect

- Shadowed and assisted the Chair in their role

2023-2025

SABER DBER-SiT Funding Co-chair

- Identified and disseminated funding sources for DBER-scholars in training

2023-2026

UNH Undergraduate Research Conference Poster Judge

- Evaluated quality of research and design of undergraduate life sciences research posters

2022-2024

UNH DBS Graduate Student Committee

- Co-founded the committee; organized graduate student seminar series; hosted outreach events

Manuscripts Reviewed

Tips, Tricks, and Techniques in the American Biology Teacher

Technical Skills

- **Programming:** R (advanced)
- **Quantitative methods:** Multilevel modeling, structural equation modeling
- **Qualitative methods:** Thematic analysis, cross-case analysis, semi-structured interview protocol development, participant observation, open-response item development
- **Reproducible research:** R Markdown, Quarto
- **Version control:** Git/GitHub

References

Melissa Aikens, Ph.D. *Melissa.Aikens@unh.edu*

Department Associate Chair, Associate Professor of Biology Education; University of New Hampshire

Easton White, Ph.D. *Easton.White@unh.edu*

Associate Professor, Department of Biological Sciences, Associate Director of Research, School of Marine Science and Ocean Engineering, University of New Hampshire

Luanna Prevost, Ph.D. *prevost@usf.edu*

Tampa Campus Associate Chair, Associate Professor, Department of Integrative Biology, University of South Florida